Project Name: OpenBank Cloud Simulation

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Objective: To containerize a full-stack banking application for consistent deployment and local development.

Backend API (backend/Dockerfile):
* https://github.com/SheBuilds05/blueMarble/blob/main/backend/Dockerfile

Frontend UI (frontend/Dockerfile):
* https://github.com/SheBuilds05/blueMarble/blob/main/frontend/Dockerfile

Service Orchestration (docker-compose.yml):
* https://github.com/SheBuilds05/blueMarble/blob/main/docker-compose.yml

Terminal Execution Log

Action: Build and launch the entire OpenBank stack.

- **Command:** `docker-compose up --build`
- **Terminal Output Snapshot:**

```
[+] Running 3/3 ⠿ Network openbank_default Created ⠿ Container
openbank-db-1 Started ⠿ Container openbank-backend-1 Started ⠿
Container openbank-frontend-1 Started
```

Action: Verify running containers.

- **Command:** `docker ps`
- **Explanation:** This confirms that the Backend, Frontend, and MongoDB instances are healthy and mapped to the correct ports.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
PS C:\Users\CAPACITI-JHB\Desktop\blueMarble> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\CAPACITI-JHB\Desktop\blueMarble>
```

Action: check Docker engine itself.

- Command: docker version
- **Explanation**: want to know whether Docker itself is alive

```
PS C:\Users\CAPACITI-JHB\Desktop\blueMarble> docker version
Client:
 Version:           29.4.0
 API version:       1.54
 Go version:        go1.26.1
 Git commit:        9d7ad9f
 Built:             Tue Apr  7 08:39:39 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.70.0 (224270)
 Engine:
  Version:          29.4.0
  API version:      1.54 (minimum version 1.40)
  Go version:       go1.26.1
  Git commit:       daa0cb7
  Built:            Tue Apr  7 08:36:03 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
PS C:\Users\CAPACITI-JHB\Desktop\blueMarble>
```

Action: Log in to the Docker Registry.

- **Command:** `docker login -u rosanovela`
- **Explanation:** This command authenticates your local terminal session with your Docker Hub account. Authentication is a required security step before you can "push" (upload) your images to a remote repository. It ensures that only authorized developers can modify or update the project images associated with your username.

```
@rosaNovela ➜/workspaces/blueMarble/backend (main) $ docker login -u rosanovela

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:

WARNING! Your credentials are stored unencrypted in '/home/codespace/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

Action: Build the Backend Docker Image.

- **Command:** `docker build -t rosanovela/bluemarble-backend:latest .`
- **Explanation:** This command instructs Docker to create a container image for the backend service.

```
@rosaNovela ➜/workspaces/blueMarble/backend (main) $ docker build -t rosanovela/bluemarble-backe
nd:latest .
 => [internal] load build definition from Dockerfile                                          0.0s
 => => transferring dockerfile: 260B                                                          0.0s
 => [internal] load metadata for docker.io/library/node:18                                    1.0s
 => [auth] library/node:pull token for registry-1.docker.io                                   0.0s
 => [internal] load .dockerignore                                                             0.0s
 => => transferring context: 2B                                                               0.0s
 => [1/7] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693  0.0s
 => => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693  0.0s
 => [internal] load build context                                                             0.0s
 => => transferring context: 1.83kB                                                           0.0s
 => CACHED [2/7] WORKDIR /app                                                                 0.0s
 => CACHED [3/7] COPY package*.json ./                                                        0.0s
 => CACHED [4/7] RUN npm install                                                              0.0s
 => CACHED [5/7] COPY . .                                                                     0.0s
 => CACHED [6/7] RUN npm run build                                                            0.0s
 => CACHED [7/7] RUN chown -R :0 /app && chmod -R g=u /app                                    0.0s
 => exporting to image                                                                        0.2s
 => => exporting layers                                                                       0.0s
 => => exporting manifest sha256:a9d404f4fbe9eb080233407065b6ff685297764a1846a0fd9d9d7cb93  0.0s
 => => exporting config sha256:ab1143850c4582c7ce987360844bc5faf59e114bbe440cb92bf32be27df  0.0s
 => => exporting attestation manifest sha256:1d9647623af11e2265d72acabd675da194d6ff5547cce  0.0s
 => => exporting manifest list sha256:315206ace3229892ff8012bb0ede22dedce0254bae88db73f85c  0.0s
 => => naming to docker.io/rosanovela/bluemarble-backend:latest                               0.0s
 => => unpacking to docker.io/rosanovela/bluemarble-backend:latest                            0.0s
```

Action: Push the Backend Image to Docker Hub.

- **Command:** `docker push rosanovela/bluemarble-backend:latest`
- **Explanation:** This command uploads the local container image to a central registry (Docker Hub). By pushing the image, you make it accessible from any other environment, such as a production server or a teammate's computer, allowing the application to be deployed anywhere without needing the original source code.

```
@rosaNovela ➜/workspaces/blueMarble/backend (main) $ docker push rosanovela/bluemarble-backend:l
atest
The push refers to repository [docker.io/rosanovela/bluemarble-backend]
3697be50c98b: Pushed
695f6d3a70f1: Pushed
b4a3bde49f66: Pushed
37927ed901b1: Pushed
cda7f44f2bdd: Pushed
461077a72fb7: Pushed
54f926098fb2: Pushed
b6bbf98f7e03: Pushed
3e6b9d1a9511: Pushed
79b2f47ad444: Pushed
e23f099911d6: Pushed
c6b30c3f1696: Pushed
26bda6f80042: Pushed
b640b745ceda: Pushed
a4b0b51afdb8: Pushed
latest: digest: sha256:315206ace3229892ff8012bb0ede22dedce0254bae88db73f85c70d063414022 size: 856
```

3. Docker Image Creation

3.1 Backend Image

The backend application was also containerized using Docker to ensure it can be deployed consistently on OpenShift.

The backend image:

- Installs backend dependencies
- Runs the application server
- Exposes the required API port

The backend is configured to run on a specific port (e.g., **8080 or 5000 depending on your setup**) and serves API endpoints that are consumed by the frontend.

The Docker image performs the following steps:

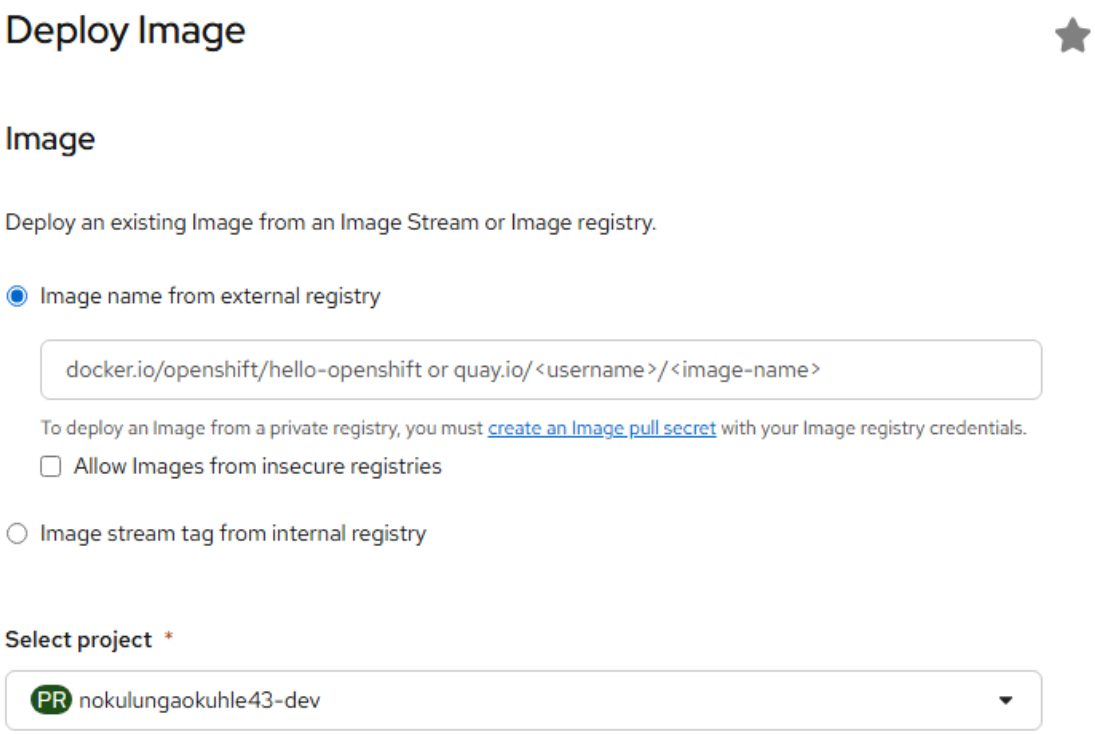
- **Build and tag image**

```
@rosaNovela ➜/workspaces/blueMarble/backend (main) $ docker build -t rosanovela/bluemarble-backe
nd:latest .
 => [internal] load build definition from Dockerfile                                          0.0s
 => => transferring dockerfile: 260B                                                          0.0s
 => [internal] load metadata for docker.io/library/node:18                                    1.0s
 => [auth] library/node:pull token for registry-1.docker.io                                   0.0s
 => [internal] load .dockerignore                                                             0.0s
 => => transferring context: 2B                                                               0.0s
 => [1/7] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693  0.0s
 => => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693  0.0s
 => [internal] load build context                                                             0.0s
 => => transferring context: 1.83kB                                                           0.0s
 => CACHED [2/7] WORKDIR /app                                                                 0.0s
 => CACHED [3/7] COPY package*.json ./                                                        0.0s
 => CACHED [4/7] RUN npm install                                                              0.0s
 => CACHED [5/7] COPY . .                                                                     0.0s
 => CACHED [6/7] RUN npm run build                                                            0.0s
 => CACHED [7/7] RUN chown -R :0 /app && chmod -R g=u /app                                    0.0s
 => exporting to image                                                                        0.2s
 => => exporting layers                                                                       0.0s
 => => exporting manifest sha256:a9d404f4fbe9eb080233407065b6ff685297764a1846a0fd9d9d7cb93  0.0s
 => => exporting config sha256:ab1143850c4582c7ce987360844bc5faf59e114bbe440cb92bf32be27df  0.0s
 => => exporting attestation manifest sha256:1d9647623af11e2265d72acabd675da194d6ff5547cce  0.0s
 => => exporting manifest list sha256:315206ace3229892ff8012bb0ede22dedce0254bae88db73f85c  0.0s
 => => naming to docker.io/rosanovela/bluemarble-backend:latest                               0.0s
 => => unpacking to docker.io/rosanovela/bluemarble-backend:latest                            0.0s
```

- **Push image**

```
@rosaNovela ➜/workspaces/blueMarble/backend (main) $ docker push rosanovela/bluemarble-backend:l
atest
The push refers to repository [docker.io/rosanovela/bluemarble-backend]
3697be50c98b: Pushed
695f6d3a70f1: Pushed
b4a3bde49f66: Pushed
37927ed901b1: Pushed
cda7f44f2bdd: Pushed
461077a72fb7: Pushed
54f926098fb2: Pushed
b6bbf98f7e03: Pushed
3e6b9d1a9511: Pushed
79b2f47ad444: Pushed
e23f099911d6: Pushed
c6b30c3f1696: Pushed
26bda6f80042: Pushed
b640b745ceda: Pushed
a4b0b51afdb8: Pushed
latest: digest: sha256:315206ace3229892ff8012bb0ede22dedce0254bae88db73f85c70d063414022 size: 856
```

- Deploy image to OpenShift



- **Deployed backend results**

Containers

Name	Image	Reso...	Ports
bluemarble-backend	image-registry.openshift-image-registry.svc:5000/nokulungaokuhle43-dev/bluemarble-backend:latest	-	8080/TCP

Conditions

Type	Status	Reason	Message
ConfigurationsReady	True	-	-
Ready	True	-	-
RoutesReady	True	-	-

- **Deployed Link**
  https://bluemarble-backend-nokulungaokuhle43-dev.apps.rm1.0a51.p1.openshiftapps.com/

3.2 Frontend Image

The frontend application was containerized using Docker to ensure consistent deployment across environments. The application was built using **React with Vite**, and a production build was generated before deployment.

The Docker image performs the following steps:

- Installs dependencies
- Builds the application using Vite
- Serves the static files using a lightweight server (`serve`)

The application is configured to run on **port 8080**, which is required for compatibility with OpenShift and Knative.

The Docker image performs the following steps:

- **Build and tag image**

```
@rosaNovela ➜/workspaces/blueMarble (main) $ cd frontend
@rosaNovela ➜/workspaces/blueMarble/frontend (main) $ docker login -u rosanovela

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:
Login Succeeded
@rosaNovela ➜/workspaces/blueMarble/frontend (main) $ docker build -t rosanovela/bluemarbl
e-frontend:v1 .
[+] Building 20.0s (9/10)                                          docker:default
 => [internal] load build definition from Dockerfile                          0.0s
 => => transferring dockerfile: 147B                                          0.0s
 => [internal] load metadata for docker.io/library/node:18                    0.7s
 => [internal] load .dockerignore                                             0.0s
 => => transferring context: 2B                                               0.0s
 => [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d  0.1s
 => => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d  0.1s
 => [internal] load build context                                             0.1s
 => => transferring context: 1.21MB                                           0.1s
 => CACHED [2/5] WORKDIR /app                                                 0.0s
 => [3/5] COPY package*.json ./                                               0.1s
 => [4/5] RUN npm install                                                    10.3s
 => [5/5] COPY . .                                                            0.2s
 => exporting to image                                                        8.5s
 => => exporting layers                                                       8.5s
 => => exporting manifest sha256:5175a2c9cf523574567a1fee9601f3ab7a738eb0ef081596d51  0.0s
```

- **Push image**

```
@rosaNovela ➜/workspaces/blueMarble/frontend (main) $ docker push rosanovela/bluemarble-fr
ontend:v1
The push refers to repository [docker.io/rosanovela/bluemarble-frontend]
dbecd6957347: Pushed
54f926098fb2: Layer already exists
328e99b839d9: Pushed
c6b30c3f1696: Layer already exists
3697be50c98b: Layer already exists
3e6b9d1a9511: Layer already exists
c9f5f5e63275: Pushed
461077a72fb7: Layer already exists
a9112e862c94: Pushed
cda7f44f2bdd: Layer already exists
37927ed901b1: Layer already exists
79b2f47ad444: Layer already exists
e23f099911d6: Layer already exists
v1: digest: sha256:1bb95342d554a8aac2d4d422ddccdbc11bb9efdf82cfa7f1c504e3a5242d0b79 size: 8
56
@rosaNovela ➜/workspaces/blueMarble/frontend (main) $
```

- **Deployed frontend results**

Status

Updating

Created at

May 6, 2026, 2:11 PM

Owner

REV bluemarble-frontend10-00001

Containers

Name	Image	Reso...	Ports
bluemarble-frontend10	image-registry.openshift-image-registry.svc:5000/nokulungaokuhle43-dev/bluemarble-frontend10@sha256:1918c9c8664f9de7e669f888c936cd8796f005eb89e42da3ea643b724993aa25	-	8080/TCP
queue-proxy	registry.redhat.io/openshift-serverless-1/kn-serving-queue-rhel9@sha256:7316b5cadfa518e05a22a11cf1ac23dad32f3b0b098b9a726f953c0e757bd9ca	-	8022/TCP, 9090/TCP, 9091/TCP, 8012/TCP, 8112/TCP

- **Deployed Link**
  Deployed Live Link